Dr Fan Zhang

Contact Information

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Research Interests

Multi-Modal Robot Manipulation, Sim-to-Real, Self-Supervised Learning, Flow Generative Models

Professional Appointments

Senior Scientist, 2025-present
Guest Scientist, 2024
Honda Research Institute EU

- Serving as Technical Lead for a team dedicated to general manipulation at Honda factories
- Accepted a Research Scientist role at Meta FAIR, 2025 (impacted by restructuring)

Eric and Wendy Schmidt AI in Science Postdoctoral Fellow, Schmidt Sciences, 2023-2024
Research Associate, 2021-2023
Imperial College London

Education

Ph.D. in Electrical and Electronic Engineering (Robotics), 2016-2020
Imperial College London
Thesis**:** Perception and Manipulation in Robotic-Assisted Dressing
Supervisor: Prof. Yiannis Demiris

Awards

Best Presentation Award, ICRA 2026 Workshop on Autonomy in Aging Societies	2026
Outstanding Paper Award, IROS 2025 Workshop on The Art of Robustness	2025
Best Research Paper (ECR), AI & Robotics Research Awards, held by UKRI TAS Hub	2025
The UK Best PhD in Robotics Award 1st place, held by Advanced Robotics @ Queen Mary	2020
Best Student Paper Award, IEEE International Conference on Mechatronics and Automation	2016

Highlighted Publications

Affordance-based Manipulation with Flow Matching
Fan Zhang, Michael Gienger
Frontiers in Robotics and AI, 2026. (paper, website, code, 270+ GitHub stars)

Learning Robot Manipulation from Audio World Models
Fan Zhang, Michael Gienger
arXiv, 2025. (paper)

Learning Garment Manipulation Policies towards Robot-Assisted Dressing
Fan Zhang, Yiannis Demiris
Science Robotics, 2022. (paper, website, code)

Probabilistic Real-Time User Posture Tracking for Personalized Robot-Assisted Dressing
Fan Zhang, Antone Cully, Yiannis Demiris
IEEE Transactions on Robotics, 2019. (paper, website, code)

Talks

Human-Centred Visual Learning Lab, University of Birmingham	2026
Intelligent Systems and Networks (ISN) Group Seminar, Imperial College London	2025
IROS Workshop on Manipulation of Deformable Objects	2025
IPAB visitor seminar, Edinburgh Centre for Robotics	2025
YorRobots Seminar, University of York	2025
Autonomous Agents Research Group, The University of Edinburgh	2025
Adaptive and Intelligent Robotics Lab, Imperial College London	2025
TAMS, University of Hamburg	2024
Social AI & Robotics Laboratory, King's College London	2023
Tsinghua University, (live audience: 150,000)	2022
Apple	2022
Human Motion Analysis for Healthcare Applications, IET	2019
The Hamlyn Centre, Imperial College London	2017

In the Press

Robotic nurse can dress a mannequin in a hospital gown, New Scientist	2022
Robotic nurse that helps you dress could aid staff shortage, Bloomberg	2019
The Times, Daily Mail, Telegraph, TechXplore	2019

Academic Activities

Inaugural Meeting - Centre for AI in Assistive Autonomy	Invited Guest
IROS 2025 Session: Factory Automation and Failure Detection	Co-chair
RSS 2024 Workshop: Learning for Assistive Robotics	Organizer
Imperial-X Breaking Topics in AI conference, Schmidt Sciences	Organizer
ICRA 2023 Workshop: Emerging Paradigms for Assistive Robotic Manipulation	Organizer
Frontiers in Robotics and AI-Robot Learning and Evolution	Review Editor
Scientific Reports, T-RO, T-RL, IJRR, ICRA, IROS, RA-L, RSS, CoRL, WACV, Humanoids	Reviewer
The Centre for AI in Assistive Autonomy, The University of Edinburgh	Collaborator
Doctoral annual review (Leonard Hinckeldey), The University of Edinburgh	Committee

Supervision & Mentorship

Leonard Hinckeldey (PhD, The University of Edinburgh, Multi-Agent RL)	2025-present
Amirreza Razmjoo (PhD, EPFL, Robot Failure Recovery)	2024-2025
Jenny Fu (PhD, Cornell University, Robot Character Learning)	2024-2025
Nikki Zhong (PhD, Imperial College London, Human Motion Modeling)	2021-2023

Research Experience

---- **General Multi-Modal Robot Manipulation**

- Serving as Technical Lead for a research team dedicated to general manipulation for Honda facilities.
- Multimodal world action model for learning robot manipulation, with special focus on contrastive latent embedding and physics learning.
- Parameter-efficient prompt tuning for learning manipulation affordance with VLM.
- Generative learning for robot manipulation with special focus on flow matching policy (code, 270+ GitHub stars).
- Supervised research: multi-agent reinforcement learning benchmark for (RLC), failure recovery with compositional diffusion models (IROS), robotic characters learning, active acoustic sensing.
- Best presentation Awards at ICRA 2026 Workshop, IROS 2025 Workshop.

---- Robot-Assisted Dressing for Bedridden Patients

- Deformable visual-tactile manipulation for garment grasping and unfolding using active learning from demonstration and model-based learning (RA-L).
- Real-to-sim-to-real garment physics learning and robot manipulation policy transfer in physics domain using contrastive learning (Science Robotics).
- Self-supervised learning for human motion modelling, including personalized user impairments modeling and dancing performance assessment (ICASSP).
- Building real and synthetic dataset of garment, including RGB-D, event images and pointcloud.
- Real-time user posture tracking using vision and haptic inputs with a probabilistic particle filter (IEEE Transactions on Robotics).
- Building controllers for hierarchical multi-task human-robot interaction (ICRA, IROS).
- Best Research Paper (Early Career Researcher), 2025 AI & Robotics Research Awards, held by UKRI TAS Hub and Responsible Ai UK.
- UK Best PhD in Robotics Award 2020 1st place, held by Advanced Robotics @ Queen Mary (ARQ).
- Collaboration with Chelsea & Westminster Hospital on robot-dressing bedridden patients.
- Live demos for National Health Service, Apple, etc.
- Media covered by Bloomberg, The Times, Daily Mail, Telegraph , NewScientist, TechXplore.
- The above works are supported by MURI Closed-Loop Multisensory Brain-Computer Interface for Enhanced Decision, TAS Trust Node UKRI Trustworthy Autonomous Systems Node in Trust, Innovate UK D-RISK Learning Edge Cases for Autonomous Vehicles.

Other Publications

Complete List (https://scholar.google.com/citations?user=X-OiekwAAAAJ&hl)

1) **Assistax: A Hardware-Accelerated Reinforcement Learning Benchmark for Assistive Robotics**
Leonard Hinckeldey, Elliot Fosong, Rimvydas Rubavicius, Elle Miller, Trevor McInroe, Fan Zhang, Patricia Wollstadt, Stefano V. Albrecht, Subramanian Ramamoorthy
Reinforcement Learning Conference (RLC), 2026. (paper, code, website)

2) **Composition of Conditional Diffusion Policies with Guided Sampling**
Amirreza Razmjoo, Sylvain Calinon, Michael Gienger, Fan Zhang
IROS, 2025. (paper, website)

3) **Generation of Real-time Robotic Emotional Expressions Learning from Human Demonstration in Mixed Reality**
Chao Wang, Michael Gienger, Fan Zhang
IROS 2025 Foundation Models for Robotic Design workshop. (paper, website)

4) **Contrastive Self-Supervised Learning for Automated Multi-Modal Dance Performance Assessment**
Yun Zhong, Fan Zhang, Yiannis Demiris
ICASSP, 2023. (paper, video, code)

5) **Visual-Tactile Learning of Garment Unfolding for Robot-Assisted Dressing**
Fan Zhang, Yiannis Demiris
RA-L, 2023. (paper, video)

6) **Learning Grasping Points for Garment Manipulation in Robot-Assisted Dressing**
Fan Zhang, Yiannis Demiris
ICRA, 2020. (paper, video)

7) **Personalized Robot-Assisted Dressing using User Modeling in Latent Spaces**
Fan Zhang, Antone Cully, Yiannis Demiris
IROS, 2017. (paper, video)